

Interpretable Dynamic Brain Network Analysis With Functional and Structural Priors

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Abstract—The dynamic functional brain network (DFBN) inherently captures topological changes in brain connectivity pattern during activity, attracting increasing attention for detecting brain disorders. However, most current DFBN analysis methods rely on data-driven modeling and ignore crucial prior knowledge of brain structure and function, resulting in weak interpretability of models. Furthermore, effectively extracting dynamic topological features from DFBN is still a challenging issue, due to its intricate spatio-temporal features coupling. In this paper, we propose an interpretable spatio-temporal tensor graph convolutional network for DFBN analysis. Firstly, by incorporating functional and structural priors into the construction of DFBN, we develop a hierarchical DFBN representation with brain region clustering that effectively captures the spatio-temporal topology among subnetworks. Secondly, we design a tensor graph convolutional network with both intra-graph propagation and inter-graph propagation to simultaneously extract the spatio-temporal features from the hierarchical DFBN. Additionally, we derive a functional subnetwork constraint to enhance the consistency within subnetworks and the differences between subnetworks, which guides the learned features to better reflect the topology prior of the brain network. Finally, self-attention is employed to fuse the learned dynamic topological features of different subnetworks for classification. Experimental results on epilepsy, ADNI and ABIDE datasets demonstrate that our method achieves competitive diagnostic performance and offers network-level interpretability for brain disease diagnosis.

Index Terms—Brain disease diagnosis, dynamic brain network, interpretability, spatio-temporal features, tensor graph convolutional network.

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I. INTRODUCTION

COMPARED to static functional brain networks, dynamic functional brain network (DFBN) inherently has the advantage in describing changes in brain connectivity topology during cognitive processes. Recently, DFBN has played a critical role in the diagnosis of brain diseases, such as epilepsy [1] and cognitive impairments [2], by providing a comprehensive understanding to investigate alterations in functional connectivity patterns. However, the complex topological structure in DFBN poses challenges for effectively extracting its spatio-temporal features. Additionally, although some data-driven DFBN analysis methods have been developed, enhancing the interpretability of the model to guide brain disease biomarker exploration remains challenging.

Most of the conventional feature extraction methods for DFBN are based on vector or non-graph-based analysis. For example, Jie et al. calculated the spatial and temporal variabilities of DFBN and integrated them for classification [2]. Wee et al. proposed estimating temporal networks by maximizing a penalized log-likelihood using a fused sparse learning algorithm [3]. Li et al. employed an adaptive ultra-weighted-lasso recursive least squares algorithm to construct DFBNs and utilized the maximum mean deviation-based autoencoder to extract spatio-temporal features [4]. In addition, time series models are also widely used to estimate DFBNs. For instance, Lindquist et al. adopted the generalized autoregressive conditional heteroscedasticity model to capture the dynamic variance and correlation of fMRI signals [5]. Vidaurre et al. used a hidden Markov model to describe the state sequence of the brain, so as to study the organization form of the brain network over time [6]. Degras et al. developed a Markov-switching state-space model to analyze the interdependence and dynamics within regimes [7]. However, these non-graph analysis methods necessitate converting the graph structure of DFBNs into vectors, potentially leading to the loss of valuable dynamic topological information in DFBNs [8].

In recent years, to preserve the dynamic topology of DFBNs, some graph preservation-based analysis methods have been proposed, primarily relying on tensor decomposition [9] and graph convolutional network (GCN) [10]. Tensor decomposition methods initially represent DFBNs as high-dimensional tensors to unveil the evolution of topology, and then seek the low-dimensional intrinsic representation of

tensors for feature extraction. For example, Zhang et al. used tensor decomposition with orthogonality and partial symmetry constraints to analyze DFBNs [9]. Zhu et al. integrated brain connectivity across time domains to construct DFBNs, and then employed tensor decomposition to extract dynamic topological features [11]. Besides, GCN and its variants that simultaneously learn topology and node features have also been employed for the analysis of DFBNs. For instance, Dahan et al. used GCN and temporal convolutions to model the topological structure and temporal flow in DFBNs [12]. Kong et al. proposed to extract spatial and temporal information from DFBNs by graph attention network and temporal fusion module, respectively [13]. It is worth noting that these methods independently extract spatial and temporal features from DFBN, making them difficult to effectively couple the inherent spatio-temporal correlation contained in the brain.

In addition, the aforementioned data-driven methods primarily focus on the prediction or classification performance of DFBNs, making it challenging to effectively incorporate prior knowledge of brain function and structure, consequently leading to weak interpretability. Neurological studies have elucidated that the brain is a highly modular system, wherein each module or subnetwork is tasked with specific functions and comprises numerous interconnected brain regions. Notably, the brain network demonstrates pronounced small-world properties. For instance, brain regions within the same subnetwork exhibit close-knit connections, while connections with brain regions spanning different subnetworks are comparatively sparse [14]. Several studies have been proposed to explore the association between brain diseases and subnetworks. For example, Wang et al. proposed a modular DFBN learning framework and investigated the impact of the central executive network, salience network and default mode network (DMN) on the brain disease diagnosis [15]. Yuan et al. designed a spatio-temporal brain attention skip network to model the personalized dynamic patterns of the DMN, and identified the abnormal connectivity in patients with mild cognitive impairment [16]. However, these DFBN analysis methods solely consider assessing the correlation between individual or limited subnetworks and brain disorders. They are hard to harness the topological information of the entire brain network, thus hindering the comprehensive understanding of the interplay between diverse subnetworks and brain diseases.

To overcome these challenges, we propose a hierarchical representation model for DFBN integrating functional and structural priors, and devise a spatio-temporal tensor graph convolutional network (ST2GCN) to collaboratively extract dynamic topological features from hierarchical DFBNs. Firstly, we use sliding windows and Pearson's correlation (PC) to construct DFBNs. Given that the pre-defined subnetwork closely aligns with the brain structure and functional organization [17], [18], it is beneficial to investigate the impact of brain diseases on various cognitive functions. Therefore, for the brain network under each window, we categorize brain regions and connections with similar functions into distinct subnetworks based on functional and structural clustering [19], encompassing DMN, fronto-parietal network (FPN), salience network (SAN), attention network (ATN), somatosensory and

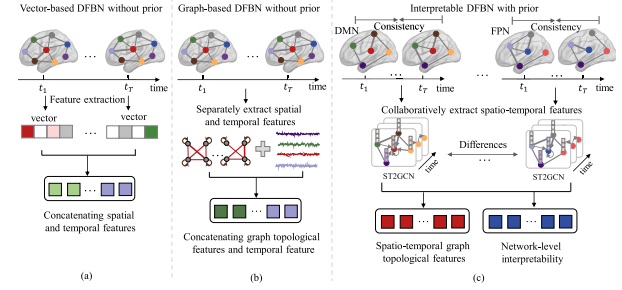


Fig. 1. The comparison between the conventional DFBNs analysis methods and the proposed method. (a) vector-based DFBN analysis methods, (b) graph-based DFBNs analysis methods, and (c) the proposed spatio-temporal graph topology collaborative analysis method with functional and structural priors.

motor network (SMN), visual network (VN), auditory network (AN), and other networks (ON) composed of brain regions and connections that are not included in the above seven subnetworks. Secondly, we introduce ST2GCN which integrates intra-graph and inter-graph propagation to collaboratively extract spatio-temporal features from dynamic functional subnetworks. The intra-graph propagation aggregates the structure information of neighboring nodes from each subnetwork, while the inter-graph propagation extracts the temporal information by connecting subnetworks under different windows to construct a series of virtual graphs. Then, to emulate the characteristics of strong coupling within subnetworks and loose coupling between subnetworks in the brain, we develop a functional subnetwork constraint to bolster the consistency of feature representation within subnetworks, and accentuate the distinctions in feature representation among subnetworks. Finally, we use the self-attention mechanism to fuse the features of different subnetworks for classification.

Figure 1 provides a concise comparison between conventional DFBN analysis methods and the proposed method. Specifically, Figure 1(a) shows the vector-based DFBNs analysis methods, which ignore extracting graph structure features among brain connectivities. Figure 1(b) denotes the graph-based DFBN analysis methods, which usually extract temporal and spatial features separately. The above two categories of methods struggle to leverage prior knowledge of brain function and structure, such as quantitatively analyzing the association between subnetworks and brain diseases. Figure 1(c) illustrates the schematic diagram of the proposed spatio-temporal graph topology analysis method. Compared with the previous methods, our approach not only offers a unified framework for effectively extracting spatio-temporal topological information within DFBNs, but also enhances model interpretability by incorporating functional and structural priors simultaneously. In brief, the proposed method has the following contributions:

(1) We propose an interpretable spatio-temporal topology collaborative extraction model for DFBN analysis. By seamlessly integrating both functional and structural priors, the proposed model facilitates decoupling the intricate interactions among subnetworks.

(2) We develop a hierarchical representation of DFBN based on structural clustering and introduce the functional subnetwork constraint to enhance consistency within subnetworks

while emphasizing differences among them, thereby aligning with functional connectivity clustering topology.

(3) To capture dynamic topological changes of subnetworks effectively, we incorporate both intra-graph propagation and inter-graph propagation into a unified framework. Intra-graph propagation is utilized to extract spatial information from subnetworks, while inter-graph propagation constructs a series of virtual graphs to extract temporal information.

(4) Experimental results on epilepsy, ADNI and ABIDE datasets demonstrate that our method surpasses state-of-the-art brain network analysis methods. Furthermore, our approach offers insights into the interpretability of functional subnetworks concerning brain disorders.

II. RELATED WORK

A. Spatio-Temporal Graph Convolution for DFBNs

There are a large number of graph structure data in real world applications, such as brain networks [20], traffic networks [21], etc. In order to capture the dynamic topological features of these graph data, some spatio-temporal graph convolution methods have been proposed. Specifically, the spatio-temporal network combines GCN and temporal convolutional network (TCN) to extract the spatial and temporal features in stages, whose general form can be expressed as follows:

$$Z = TCN(GCN(A, X)) \quad (1)$$

where A and X denote the adjacency matrix and node feature of the graph respectively. For example, Gadgil et al. constructed DFBNs to model the non-stationary nature of functional connectivity, and used GCN and 1D convolutional neural network to obtain spatio-temporal characteristics [22]. Xu et al. used multi-channel GCN to extract the structural and functional features of DFBNs, and utilized long short-term memory to simulate the interactions of brain regions across windows [8]. Kim et al. first employed the graph isomorphism network to fuse structural and node features, and then used spatial and temporal attention mechanisms to automatically focus on the spatio-temporal information in the DFBNs [20].

However, these dynamic graph analysis methods extract temporal and spatial features independently, which makes them difficult to effectively capture the tightly coupled spatio-temporal topological properties in brain networks. In this paper, we propose a collaborative extraction method for spatio-temporal graph features, incorporating both intra-graph and inter-graph propagation. Intra-graph propagation aggregates spatial structure information from neighboring nodes within subnetworks, while inter-graph propagation captures temporal information of subnetwork by constructing virtual graphs.

B. Interpretability Modeling on Brain Networks

Given the occurrence of brain diseases leading to abnormal connectivities of brain networks [2], leveraging interpretability modeling on these networks shows promise in identifying biomarkers closely associated with these diseases. For example, Li et al. proposed the Regions of Interest (ROI)-aware

GCN to leverage the functional and topological information of the brain network, and adopted a special pooling to emphasize these important ROIs [23]. Cui et al. developed an interpretable framework to explore disorder-specific ROIs and prominent connections [24]. Kan et al. considered the learnable generation of brain networks and studied the explainability of the generated brain network towards downstream tasks [25]. However, these methods primarily focus on the differences among brain regions and do not reveal the association of different subnetworks with brain diseases. Integrating the hierarchical functional and structural priors from the brain region level to the subnetwork level helps to describe the brain's modular organization, thereby offering subnetwork-level interpretability for brain disease diagnosis.

In physiology, the brain is composed of subnetworks responsible for different functions, forming the neurobiological basis of information processing and functional integration [26]. Several studies have applied functional subnetworks in brain network analysis. For example, Bannadabhavi et al. [27] and Kan et al. [28] explored the association between brain diseases and multiple functional subnetworks. Ghayem et al. first employed independent component analysis to reveal interactions between subnetworks, then incorporated dictionary learning to identify key subnetwork connections [29]. In this paper, we explicitly model the modular attributes of tight coupling within subnetworks and loose coupling between subnetworks based on prior knowledge. Moreover, we design the ST2GCN to collaboratively capture the dynamic topological changes of subnetworks, thereby comprehensively exploring the interpretability between brain diseases and the spatio-temporal characteristics of subnetworks.

III. PROPOSED METHOD

In this paper, we propose an interpretable spatio-temporal graph convolutional network with functional and structural prior, which not only effectively captures the topological changes in DFBNs, but also inherently possesses interpretability at the subnetwork level. The overall architecture of the proposed method is illustrated in Figure 2. We describe the proposed method in detail below.

A. Spatio-Temporal Tensor Graph Convolutional Network

1) *Construct Dynamic Functional Subnetworks Based on Structural Clustering*: We assume that the rs-fMRI time series data for each subject is $X = (x_1, x_2, \dots, x_N)^T \in R^{N \times M}$, where N is the number of brain regions, M represents the number of collected successive time signals. In order to effectively characterize the spatio-temporal evolution of brain networks, we first construct the DFBN. The changes in brain networks are usually progressive, and wide-spaced non-overlapping windows are beneficial for capturing more significant dynamics [30], [31]. Specifically, we partition the fMRI signals using non-overlapping sliding windows to obtain T sub-signals $F = (F_1, F_2, \dots, F_T) \in R^{T \times N \times S}$, where $F_t = (x_1(t), x_2(t), \dots, x_s(t)) \in R^{N \times S}$ ($t = 1, 2, \dots, T$), and $S = M/T$ is the length of the sub-signals. Next, we adopt Pearson's correlation to construct the functional brain network A_t for

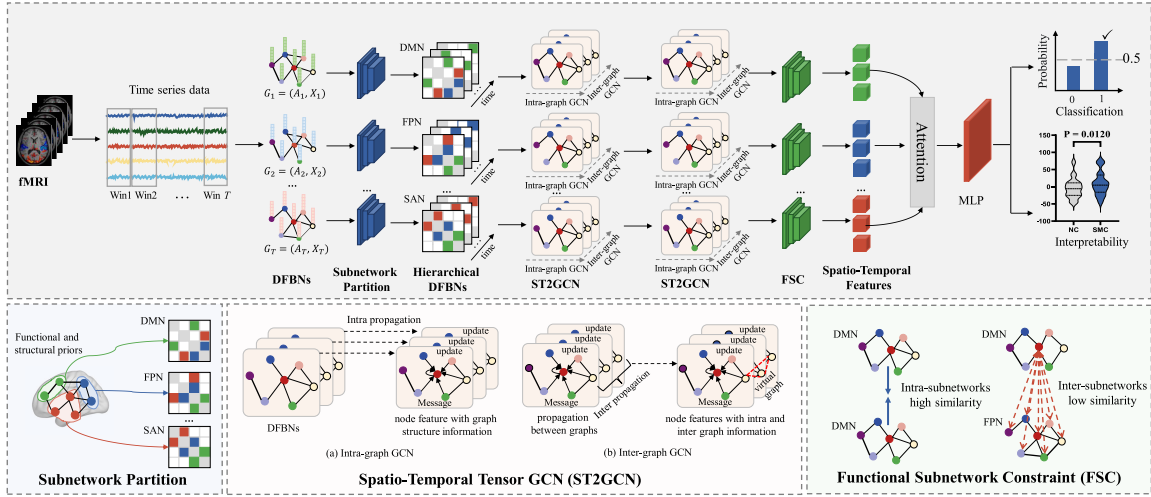


Fig. 2. The proposed framework for brain disease diagnosis with interpretable ST2GCN.

each sub-signals F_t , and then the DFBNs can be represented as $A = (A_1, A_2, \dots, A_T) \in \mathbb{R}^{T \times N \times N}$. To maintain the sparsity of the brain network and mitigate noisy connections, we set the elements of the functional connectivity matrix that are below the threshold α to zero [32]. Next, we partition the functional networks within each window into C subnetworks according to pre-defined functional and structural clustering [19]. To capture the evolving patterns of brain connectivity, we store the same subnetwork under all windows as a 3D tensor. As a result, we can obtain the C dynamic functional subnetworks $\{G^c | G^c = (A^c, F^c)\}_{c=1}^C$, $A^c \in \mathbb{R}^{T \times N^c \times N^c}$, $F^c \in \mathbb{R}^{T \times N^c \times S}$, where A^c , F^c and N^c denote the adjacency matrix, node feature and the number of nodes in G^c , respectively.

2) ST2GCN: The conventional GCN struggles with handling dynamic graphs, resulting in an inability to effectively integrate spatio-temporal features in dynamic functional subnetworks. To address this limitation, we design ST2GCN with intra-graph propagation and inter-graph propagation, which can directly perform convolutional learning on dynamic tensor graphs to cooperatively capture spatio-temporal correlations. For l^{th} layer of the ST2GCN, we first perform intra-graph propagation followed by inter-graph propagation:

$$ST2GCN(A^c, F^{c(l)}) = \{F^{c(l)} \xrightarrow{f_{intra}} F_{intra}^{c(l)} \xrightarrow{f_{inter}} F^{c(l+1)}\} \quad (2)$$

where $F^{c(l)} \in \mathbb{R}^{T \times N^c \times d_l}$ is the hidden feature tensor of the l^{th} layer in ST2GCN, d_l denotes the feature dimension, $F_{intra}^{c(l)}$ represents the output feature tensor after performing the intra-graph propagation, f_{intra} and f_{inter} are the intra-graph propagation and inter-graph propagation, respectively.

a) Intra-graph propagation: The intra-graph propagation aggregates the feature of neighboring nodes for each node within the dynamic functional subnetworks. Given the adjacency tensor $A^c \in \mathbb{R}^{T \times N^c \times N^c}$, and feature tensor $F^c \in \mathbb{R}^{T \times N^c \times S}$, the output feature of the t^{th} graph in the l^{th} layer is updated according to the intra-graph propagation f_{intra} :

$$F_{intra}^{c(l)}(t, :, :) = \sigma(\hat{A}^c(t, :, :) F^{c(l)}(t, :, :) W_{intra}^{c(l,t)}) \quad (3)$$

where $\hat{A}^c(t, :, :) = \tilde{D}^c(t, :, :)^{-\frac{1}{2}} \tilde{A}^c(t, :, :) \tilde{D}^c(t, :, :)^{-\frac{1}{2}} \in \mathbb{R}^{T \times N^c \times N^c}$ is the t^{th} graph after symmetrical normalization, $\tilde{A}^c = A^c + I^c$, where I^c represents the identity tensor consisting of T identity matrices, and $\tilde{D}^c$ is the node degree tensor with T diagonal node degree matrices. In addition, the trainable weight matrix is designed to layer-specific and graph-specific, $W_{intra}^{c(l,t)}$ represents the weight matrix of the t^{th} graph at the l^{th} layer, and σ is activation function. Different from the traditional GCN, we perform graph convolution operation on the functional subnetworks under each window, thus forming a tensor version of GCN, which is more beneficial to aggregate the spatial topological features of brain networks.

b) Inter-graph propagation: Inter-graph propagation aggregates the temporal feature of the dynamic functional subnetworks G^c under different windows. For this purpose, we construct a series of virtual graphs by connecting the same nodes in different temporal graphs. Specifically, the dynamic functional subnetwork G^c actually shares a set of identical nodes $V^c = (V_1^c, V_2^c, \dots, V_{N^c}^c)$, so we construct a series of virtual graphs by connecting the same nodes $V_n^c (n = 1, 2, \dots, N^c)$ in G^c . Since all nodes and edges in the virtual graph are equally important, the connection weight between pairs of nodes in the virtual graph is set to 1. Because G^c contains N^c brain regions, we can obtain N^c virtual graphs. Considering that the virtual graph is identical for all nodes, the adjacency matrix of each virtual graph can be simply represented as the all-ones matrix $E \in \mathbb{R}^{T \times T}$. The inter-graph information propagation f_{inter} on the virtual graph is implemented by Eq. (4):

$$F^{c(l+1)}(:, n, :) = \sigma(E F_{intra}^{c(l)}(:, n, :) W_{inter}^{c(l,n)}) \quad (4)$$

where $F^{c(l+1)} \in \mathbb{R}^{T \times N^c \times d_{l+1}}$ is not only the output feature of information propagation between virtual graphs, but also the input feature of the $l+1^{th}$ layer ST2GCN. $W_{inter}^{c(l,n)}$ represents the learnable parameters during inter-graph propagation. Therefore, ST2GCN can be represented as follows:

$$F^{c(l+1)} = ST2GCN(A^c, F^{c(l)}) \quad (5)$$

ST2GCN processes the dynamic functional subnetworks G^c through intra-graph propagation and inter-graph propagation, which cannot only extract the topological information within windows, but also extract the temporal information across windows in the G^c , so as to effectively express the inherent spatio-temporal correlation in the dynamic functional subnetwork. Next, we perform the ST2GCN operation on each dynamic functional subnetwork to extract spatio-temporal features by: $\{F^{c(l+1)} = ST2GCN(A^c, F^{c(l)})\}_{c=1}^C$. It is worth noting that ST2GCN significantly differs from multi-channel GCN. Multi-channel GCN does not consider information propagation across the time dimension, potentially leading to the loss of important temporal features, whereas our method simultaneously considers both temporal and spatial dimensions in graph feature extraction.

B. Functional Subnetwork Constraint

The brain is a modular system composed of multiple subnetworks, with each subnetwork consisting of a large number of brain regions and brain connections with specific functions. Brain regions within a subnetwork are always tightly connected, while most of those located in different subnetwork are sparsely connected [33]. Based on this prior knowledge, we design the brain functional subnetwork constraint and embed it into our spatio-temporal tensor graph convolution network. It encourages the feature representation of brain regions within a subnetwork to be aggregated, while promoting discriminative feature representations for brain regions across different subnetworks. The schematic of the proposed functional subnetworks constraint is shown in Figure 3.

1) *Intra-Subnetwork Constraint*: We use cosine similarity [34] to measure the similarity of spatio-temporal feature representations of brain regions within the subnetwork and prompt this similarity to increase as training progresses:

$$L_{M_1} = - \sum_{t=1}^T \sum_{c=1}^C \sum_{i,j=1}^{N_c} \frac{(F_t^c)_i \cdot (F_t^c)_j}{\|(F_t^c)_i\| \cdot \|(F_t^c)_j\|} \quad (6)$$

where $(F_t^c)_i$ and $(F_t^c)_j$ are the feature embeddings of the i^{th} and j^{th} nodes in the c^{th} dynamic functional subnetworks at t^{th} time window, $\cdot$ denotes inner product operation, and $\|\cdot\|$ denotes the Frobenius norm.

2) *Inter-Subnetwork Constraint*: In order to encourage the diversity of features between dynamic functional subnetworks, we adopt Euclidean distance [35] to measure the difference of feature representations between different subnetworks, and guide this difference to continuously increase as training progresses. Since different dynamic subnetworks contain varying numbers of nodes, the dimensions of the extracted spatio-temporal features differ accordingly. Therefore, we first use full connection (FC) layer to align the feature dimensions of multiple subnetworks: $H_c = FC(Flatten(F^c))$, where F^c represents the feature embeddings in the c^{th} dynamic functional subnetworks, $Flatten(*)$ is the flatten operation, $H^c \in R^d$ represents the aligned feature embedding, and d is the embedding dimension. After that, we calculate the Euclidean distance between feature embeddings of different

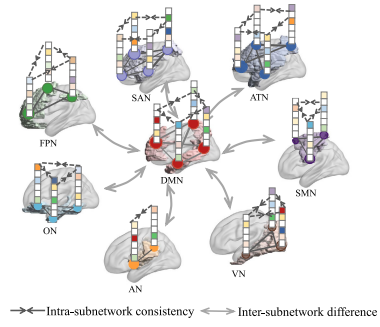


Fig. 3. Schematic of the functional subnetwork constraints. Specifically, we use DMN as an example to describe the difference in feature representation between functional modules. The different color coatings in the brain indicate the distributed location of the subnetwork. Among them, the features of brain regions belonging to the same functional subnetwork are similar.

dynamic functional subnetworks:

$$L_{M_2} = - \sum_{c_1, c_2=1}^C \|H_{c_1} - H_{c_2}\|_2^2 \quad (7)$$

C. Attention Mechanism

Considering that each functional subnetwork contributes differently to the brain network representation, we introduce a self-attention mechanism to form a whole-brain level representation by weighted combination of features from different subnetworks. Specifically, we first pack the spatio-temporal features of different subnetworks into a matrix $Z \in R^{C \times d}$. Then, we perform linear mappings on matrix Z , including $Q = ZW_q$, $K = ZW_k$ and $V = ZW_v$, where W_q , W_k and W_v all denote learnable weight matrices. Next, we capture the self-attention coefficient between functional subnetworks by calculating the scaled dot-product correlation between Q and K . Therefore, we can calculate the self-attention by mapping Q and a set of K - V pairs, and then obtain the whole brain network embedding vector H :

$$H = \frac{1}{C} \text{softmax}(QK^T / \sqrt{d})V \quad (8)$$

where softmax is used to normalize the attention weight.

Finally, we fed the comprehensive brain network embedding into multilayer perceptron to predict the diagnostic result. The overall objective function of the proposed model can be formulated as:

$$L = L_{CE} + \lambda_1 L_{M_1} + \lambda_2 L_{M_2} \quad (9)$$

where L_{CE} denotes cross-entropy loss, λ_1 and λ_2 are the hyperparameters that determine the relative importance of intra-subnetwork and inter-subnetwork constraint loss items.

IV. EXPERIMENT

A. Materials

1) *Data Acquisition*: In this paper, we conduct experiments on the collected epilepsy dataset, the public ADNI dataset (<http://adni.loni.usc.edu/>) and the public ABIDE dataset (https://fcon_1000.projects.nitrc.org/indi/abide/). The three datasets are described as follows: Epilepsy: The fMRI data was collected from Jinling Hospital, Nanjing University

School of Medicine. The dataset contains a total of 306 subjects, including 114 normal controls (NC), 103 frontal lobe epilepsy (FLE) patients, and 89 temporal lobe epilepsy (TLE) patients. The equipment for collecting data is the Siemens Trio 3T scanner. The scan parameters of fMRI are as follows: repetition time = 2000 ms, echo time = 30 ms, flip angle = 90° , and voxel size = $3.75 \times 3.75 \times 3.75 \text{ mm}^3$. ADNI: We downloaded fMRI data for 226 samples from ADNI2 and ADNI3 datasets. The ADNI dataset contains 82 normal controls (NC), 76 significant memory concern (SMC) patients, and 69 early mild cognitive impairment (EMCI) patients. Among them, the NC group contains 40 males and 42 females with an average age of 74.31. The SMC group consists of 50 males and 29 females with a mean age of 77.24, and the EMCI group consists of 37 males and 32 females with a mean age of 74.77. ABIDE: We obtained a total of 630 subjects' fMRI data from the ABIDE dataset, including 337 NC and 293 autism spectrum disorder (ASD) patients. Among them, the NC group contains 267 males and 70 females with an average age of 16.59, and the ASD group consists of 255 males and 38 females with an average age of 17.44.

2) Data Preprocessing: We use the same way to preprocess the epilepsy and ADNI datasets. Specifically, we use the SPM8 as implemented in the DPARSF toolbox [36] for all fMRI data preprocessing. To correct and realign the initial images, the serialized data is split into several pieces and then adjusted to the EPI template. We utilize detrending to reduce the influence of head motion and the interference from CSF and white matter. Then, we used the Automated Anatomical Labeling (AAL) atlas to divide the fMRI of the epilepsy dataset into 90 ROIs with 240 time points, and the AAL atlas was used to divide fMRI of the ADNI dataset into 90 ROIs with 197 time points. For the ABIDE dataset, preprocessing operations included slice time correction and motion correction. Next, we used the AAL atlas to divide this dataset into 90 ROIs, and adopted the slicing operation to uniformly process the data from different centers into fMRI signals with 176 time points.

B. Comparison Methods

To evaluate the effectiveness of the proposed method, we compare it with 16 brain network analysis methods, consisting of ten static brain network analysis methods, and six DFBNs analysis methods. Static brain network analysis methods include a logistic regression model based on functional connectivity (FC-LR) [37], L2 regularized regression model based on functional connectivity (FC-L2LR) [38], community-aware brain transformer (Com-BrainTF) [27], GCN [10], graph attention networks (GAT) [39], brain graph neural network (BrainGNN) [23], interpretable brain graph neural network (IBGNN) [24], functional brain network generation (Fbnetgen) [25], brain network transformer (BNT) [40], and dictionary learning-based brain network classification and analysis (DL-BNCA) [29]. DFBNs analysis methods include dynamic brain transformer (DRAT) [28], hidden Markov model (HMM) [6], Markov-switching state-space models (MS2M) [7], spatio-temporal fMRI (ST-fMRI) [12], spatio-temporal graph convolution network (ST-GCN)

[22] and spatio-temporal attention graph isomorphism network (STAGIN) [20].

C. Experimental Settings

The proposed method is implemented in PyTorch using an NVIDIA GeForce RTX 3080 GPU. For each dynamic functional subnetwork, we use two-layer ST2GCN to extract spatio-temporal features. Then, the single-head self-attention mechanism is used to fuse the features of different subnetworks to obtain the feature representation of the global brain network. We then pull the global features into a one-dimensional vector and use MLP to predict the class of each individual. Here, the number of neurons in the two fully connected layers is 32 and 2, respectively. In addition, we set dropout to 0.5 to avoid overfitting. We optimize the proposed method via the Adam algorithm [41], with the learning rate of $5e-4$. The number of epochs is set to 200, and the batch size is set to 20. The objective function of the proposed method is a composite loss function incorporating three distinct loss terms. To facilitate model convergence, we implement the following experimental setup. Firstly, we take the logarithm of both the cosine similarity loss and the Euclidean distance loss to ensure that the three types of losses are on the same scale. Secondly, we set the hyperparameters λ_1 and λ_2 to encourage the model to comprehensively consider the impact of multiple losses, ensuring that the gradients descend stably. It is worth noting that we also implement the gradient clipping strategy to prevent gradient explosion, thereby improving the convergence speed of the model.

In the experiment, we evaluate the performance of the proposed method and comparison methods on three datasets by 10-fold cross-validation. Each dataset is divided into 10 mutually exclusive subsets of approximately the same size, nine of which are used as the training set and the remaining one is treated as the test set. We divide 20% of the samples in the training set into the validation set, and the model parameters that perform best on the validation set are used for testing on the test set. The experiment is repeated a total of ten times, and we report the average results and the standard deviation of the ten experiments. Our code is available at: <https://github.com/xbrainnet/ST2GCN.git>.

We use grid search to determine the optimal hyperparameters in the model, such as the number of sliding windows T , brain connectivity threshold α , λ_1 and λ_2 . Specifically, the number of sliding windows T is set from $\{1, 2, 3, 4, 5, 6, 7, 8\}$, and α ranges from 0.1 to 0.9 with a step size of 0.1. λ_1 and λ_2 are tuned within the range of $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 1, 10\}$. These hyperparameter combinations are placed in an outer loop of 10-fold cross-validation to find the best performance based on cross-validation. Specifically, T is set to 5, and α is set to 0.6. For the epilepsy dataset, λ_1 is set to $1e-4$ while λ_2 is set to $1e-2$. For the ADNI dataset, λ_1 is set to $1e-4$ while λ_2 is set to $1e-1$. For the ABIDE dataset, both λ_1 and λ_2 are specified as $1e-4$. We evaluate the performance of the proposed method using four binary classification tasks on the epilepsy dataset, including NC vs. FLE&TLE, NC vs. FLE, NC vs. TLE and FLE vs. TLE. We also use four binary classification tasks on the ADNI dataset to evaluate the performance of the pro-

TABLE I

THE CLASSIFICATION PERFORMANCE (MEAN/STD) OF THE PROPOSED METHOD AND COMPARISON METHODS ON THE EPILEPSY DATASET (%)

Method	NC vs. TLE&FLE			NC vs. FLE			NC vs. TLE			TLE vs. FLE		
	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC
FC-LR	68.3/06.8	77.1/05.0	71.7/07.3	68.0/08.4	61.1/24.5	61.0/10.1	67.5/08.1	65.0/10.6	66.2/16.7	61.5/08.4	64.6/10.0	66.6/06.1
FC-L2LR	67.8/05.6	74.8/04.5	76.2/08.2	68.5/08.4	65.7/16.1	64.4/15.1	71.5/08.4	67.3/08.0	71.1/11.6	57.3/07.5	60.1/08.2	56.0/11.4
Com-BrainTF	78.8/05.4	82.1/06.7	79.2/07.4	78.1/07.5	56.4/33.6	72.8/18.5	81.3/08.8	69.5/25.1	80.6/12.0	73.1/06.3	68.4/24.0	59.6/16.0
GCN	69.3/06.5	76.1/06.3	61.8/11.7	69.5/05.2	56.4/03.0	61.6/06.9	71.8/08.6	53.5/07.0	69.4/07.5	67.2/06.3	67.5/01.5	56.8/01.5
GAT	71.3/09.6	80.8/07.8	66.6/09.7	69.1/05.3	62.5/09.8	68.7/05.6	66.8/05.7	54.5/04.9	64.8/04.6	64.6/07.7	67.0/04.1	67.4/07.8
BrainGNN	75.5/07.3	81.0/08.1	76.1/09.9	73.7/05.5	70.7/08.9	70.8/07.6	78.3/06.1	70.3/09.6	79.8/07.5	67.2/07.0	69.7/08.0	61.4/07.2
IBGNN	76.9/03.3	80.4/03.7	67.3/09.2	76.8/08.1	67.0/07.7	72.5/07.2	80.7/05.2	75.0/06.6	82.4/09.3	70.3/05.1	70.0/08.9	65.3/08.9
Fbnetgen	71.5/05.6	76.0/05.8	65.7/09.8	71.0/06.8	66.0/09.4	68.9/09.1	72.0/07.8	59.1/22.0	63.6/12.6	65.1/06.6	70.2/09.2	60.4/09.7
BNT	79.8/05.6	83.3/05.4	78.8/07.1	77.4/07.7	74.1/09.9	76.3/05.8	81.1/07.1	76.7/09.8	83.9/08.4	69.8/05.1	73.8/05.4	67.6/08.9
DL-BNCA	68.3/03.2	79.7/01.8	76.8/09.9	65.8/12.7	58.0/17.0	74.0/12.3	71.2/05.5	57.7/08.5	80.0/06.4	55.2/08.7	67.3/06.4	50.4/20.8
DRAT	77.1/05.4	81.6/05.0	78.7/04.5	79.4/06.3	77.0/11.0	78.8/08.0	82.5/06.1	78.5/04.8	81.9/09.3	70.6/08.9	74.6/08.2	67.7/13.1
HMM	64.3/06.8	71.1/08.0	55.9/07.9	70.0/08.3	63.9/14.4	64.9/13.5	64.0/04.9	51.5/18.0	56.4/10.0	67.2/08.0	65.1/14.7	54.0/11.2
MS2M	69.6/05.7	76.8/04.7	73.5/06.3	62.1/09.2	61.0/12.8	68.0/07.4	66.8/09.4	60.3/12.9	76.0/12.1	58.3/09.3	63.3/08.3	62.4/12.5
ST-fMRI	77.1/05.6	80.2/08.5	69.9/10.3	75.1/07.5	71.9/08.7	68.4/06.3	75.7/05.4	71.4/08.8	70.7/07.2	66.2/06.6	70.0/09.4	59.9/06.8
ST-GCN	79.6/09.6	82.8/08.9	80.9/07.3	76.0/03.7	73.1/04.8	74.3/09.1	80.2/05.4	73.8/06.1	78.2/05.9	67.4/05.8	77.9/05.7	60.5/08.4
STAGIN	79.7/06.6	82.9/07.4	69.9/06.9	75.1/08.8	69.6/09.8	67.5/09.6	78.6/05.2	72.9/07.7	77.9/05.5	67.2/06.2	68.6/06.2	56.8/06.7
Ours	84.0/05.0	88.1/04.4	82.0/06.7	83.5/05.5	80.2/05.5	87.1/06.0	84.1/05.9	81.8/08.4	82.4/08.5	76.9/05.7	79.0/09.4	74.2/09.1

TABLE II

THE CLASSIFICATION PERFORMANCE (MEAN/STD) OF THE PROPOSED METHOD AND COMPARISON METHODS ON THE ADNI DATASET (%)

Method	NC vs. SMC&EMCI			NC vs. SMC			NC vs. EMCI			SMC vs. EMCI		
	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1	AUC
FC-LR	72.5/07.4	79.4/04.9	81.3/07.3	83.4/08.6	79.8/12.7	85.2/10.8	78.9/07.5	74.0/11.1	72.9/15.8	54.6/10.6	59.1/10.6	58.9/16.1
FC-L2LR	80.8/10.0	84.2/09.7	86.0/11.6	86.1/11.7	83.0/17.6	85.3/15.9	81.3/07.6	77.8/07.6	83.8/10.7	61.5/10.3	64.1/12.0	61.0/15.1
Com-BrainTF	76.1/05.7	88.6/05.4	89.1/06.1	90.0/07.3	88.0/09.6	88.1/10.3	87.5/09.3	86.4/10.0	89.4/09.7	75.8/05.8	74.7/10.6	69.0/12.4
GCN	79.6/09.7	82.9/08.7	72.6/15.0	85.5/06.8	83.4/07.6	84.0/05.9	85.3/04.0	81.2/07.3	85.2/09.9	75.1/05.3	74.9/09.6	67.9/02.9
GAT	81.0/05.5	64.9/08.8	76.8/07.5	84.9/06.9	83.1/09.8	85.4/09.9	83.3/05.8	78.6/08.9	86.7/09.2	76.4/08.2	76.0/09.8	74.5/10.0
BrainGNN	84.9/04.5	87.0/06.0	88.3/03.5	84.2/07.2	82.3/08.7	86.3/05.0	88.0/05.0	84.9/08.5	91.0/07.7	77.0/06.6	77.6/06.9	75.8/09.9
IBGNN	86.3/04.8	87.9/07.8	85.6/06.0	86.8/07.6	85.3/09.1	83.6/09.5	88.7/07.3	85.7/09.3	85.7/09.5	86.8/04.8	85.8/07.8	88.3/06.8
Fbnetgen	81.5/07.6	83.5/09.0	81.7/09.4	84.9/04.0	79.6/07.9	81.8/10.0	84.8/05.9	81.9/08.4	85.1/08.0	74.6/06.6	75.4/05.9	75.9/09.2
BNT	86.3/06.5	89.7/03.9	89.7/04.6	86.3/08.8	84.4/10.0	88.5/08.8	86.0/05.5	83.9/05.4	79.8/09.4	86.1/07.1	86.7/05.6	88.0/09.2
DL-BNCA	81.3/07.2	84.6/06.8	89.0/07.7	83.2/09.7	81.9/11.6	86.7/11.2	88.5/06.2	80.7/07.0	92.4/04.6	60.7/10.0	57.2/11.5	67.6/11.7
DRAT	86.3/06.7	88.3/06.1	89.2/07.7	87.5/11.2	83.0/17.5	91.1/14.7	87.5/07.9	82.0/12.2	87.8/09.9	80.8/08.4	79.4/12.8	73.5/14.1
HMM	83.9/06.3	86.4/06.0	82.8/08.1	84.2/12.1	77.4/27.5	89.5/12.1	80.0/10.7	79.6/10.5	77.8/12.7	74.2/17.7	72.5/19.9	55.1/26.3
MS2M	77.3/14.0	83.4/09.7	87.9/12.8	79.1/08.4	78.2/11.3	88.9/12.3	78.3/07.9	71.0/11.2	85.3/07.1	58.5/13.0	66.8/11.9	67.9/15.1
ST-fMRI	86.3/06.5	89.4/05.5	89.0/06.4	87.9/04.6	85.4/06.4	87.7/06.6	88.0/08.1	86.7/08.6	88.7/09.8	85.4/07.1	82.8/08.8	86.2/06.5
ST-GCN	85.8/07.3	87.9/07.3	88.5/09.4	87.4/07.3	86.4/06.9	87.4/09.7	87.3/07.0	86.2/08.1	86.9/08.8	85.4/05.7	85.6/05.4	87.1/06.3
STAGIN	86.3/06.3	89.3/05.4	86.4/10.0	89.3/07.5	87.4/08.2	87.9/09.0	90.7/06.8	87.7/09.4	87.4/09.5	87.0/07.6	86.8/07.1	86.8/09.0
Ours	93.4/02.2	95.1/02.3	92.5/06.5	93.7/05.0	93.0/05.1	95.3/04.8	95.3/07.3	94.0/09.2	94.3/08.8	91.0/05.6	90.7/06.5	92.0/07.1

posed method, including NC vs. SMC&EMCI, NC vs. SMC, NC vs. EMCI and SMC vs. EMCI. In addition, we assess the method's performance on the ABIDE dataset through a binary classification task: NC vs. ASD. The performance of each classification task is evaluated by accuracy (ACC), F-Measure (F1) and the area under the receiver operation characteristic curve (AUC).

D. Classification Performance

Table I, Table II and Table III display the experimental results of the different methods on epilepsy, ADNI and ABIDE datasets, respectively. From these tables, we can find that the proposed method achieves the optimal diagnostic performance among all methods on three datasets. In addition, we plot the ROC curves of the proposed method and the comparison methods on the four tasks NC vs. FLE, NC vs. TLE, NC vs. SMC, and NC vs. EMCI in Figure 4. As can be seen in Figure 4, the ROC curve of the proposed method is located in the upper left corner of all the comparison methods, which verifies that

this method has the highest AUC value. These experimental results demonstrate the effectiveness of the proposed method in diagnosing brain diseases. The reason why our method performs better is that it not only considers the consistency within subnetworks and the differences between subnetworks, but also collaboratively extracts the spatio-temporal features of dynamic functional subnetworks. Compared with FC-LR, Com-BrainTF, GCN and GAT, which can only capture the static topology of the brain network, our method can capture the dynamic information of the brain. Compared with BrainGNN, IBGNN, FBNetGen and BNT, which ignore the prior knowledge in the brain, our method emphasizes the similarities within functional subnetworks while enhancing the differences between subnetworks, aligning more closely with the functional dynamics of the brain. Different from ST-fMRI, ST-GCN and STAGIN, which extract the spatio-temporal features of DFBNs in stages, the proposed method can collaboratively extract the spatio-temporal feature of DFBNs, which is beneficial to capture the latent dynamic topological information of DFBNs. Furthermore, during the grid search for

TABLE III

THE CLASSIFICATION PERFORMANCE (MEAN/STD) OF THE PROPOSED METHOD AND COMPARISON METHODS ON THE ABIDE DATASET (%)

Method	NC vs. ASD		
	ACC	F1	AUC
FC-LR	58.7/05.4	51.2/06.4	60.9/06.6
FC-L2LR	61.9/05.6	59.6/07.1	66.5/05.9
Com-BrainTF	69.7/04.5	67.4/08.4	66.0/07.1
GCN	60.3/04.4	54.5/07.3	57.3/07.2
GAT	70.0/04.4	63.8/08.2	72.0/05.9
BrainGNN	72.5/03.2	67.6/08.4	74.4/05.2
IBGNN	63.7/03.1	48.1/25.1	58.9/08.1
Fbnetgen	70.2/04.8	65.5/08.8	64.3/09.6
BNT	72.5/04.4	66.7/08.3	73.8/05.2
DL-BNCA	53.5/02.5	24.3/06.4	55.4/03.1
DRAT	68.2/04.4	56.8/15.7	69.6/06.4
HMM	66.2/04.9	62.9/09.3	67.8/07.9
MS2M	63.8/03.7	54.5/14.6	61.0/04.5
ST-fMRI	71.6/04.5	64.3/15.2	64.2/08.2
ST-GCN	67.5/05.6	65.8/06.8	52.0/10.1
STAGIN	67.0/03.6	62.8/06.5	67.1/06.2
Ours	73.8/03.5	68.0/10.0	75.16/03.3

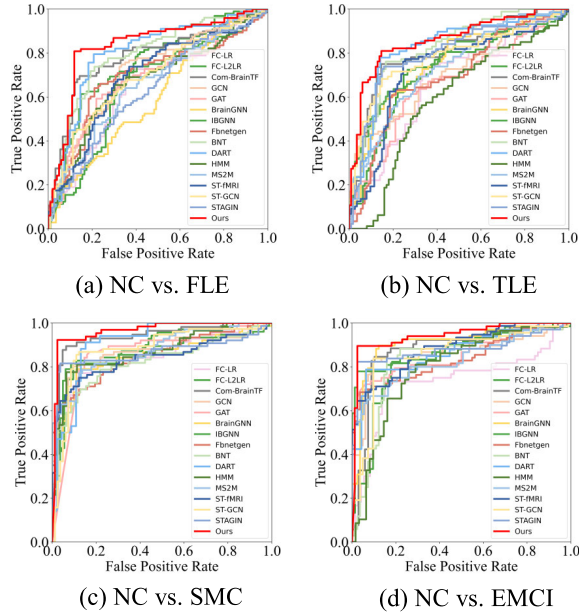


Fig. 4. ROC curves of the proposed method and comparison methods for brain disease diagnosis on different classification tasks. A larger area under the ROC curve of a method indicates better performance.

the optimal number of sliding windows T , we find that DFBN exhibits superior classification performance compared to the static brain network ($T = 1$). This is because DFBN contains more complex time-varying topological features, which can effectively reflect the influence of brain diseases on the brain's spatio-temporal characteristics.

To assess the proposed method's efficacy in multi-disease diagnostics, we perform three-class classification experiments on the epilepsy and ADNI datasets. For the NC vs. FLE vs. TLE task, the ACC, F1 and AUC are 66.00%, 58.18% and 74.77%, respectively. For the NC vs. SMC vs. EMCI task, the ACC, F1 and AUC are 72.43%, 69.37% and 79.19%, respectively. Compared to binary classification tasks, the performance of the proposed method and the comparison methods on the three-class classification task is somewhat reduced. This

may be due to the more complex feature distribution in the three-class classification task. In order to effectively improve the accuracy of multi-disease diagnosis, we will enhance the proposed method in future research to better distinguish features from different categories.

We also explore the computational resources required by the proposed method. Experimental results indicate that the model requires only 0.0433 million model parameters and 0.2039 million floating-point operations. Additionally, it takes just 1.9670 seconds to run one epoch. Therefore, our method can achieve superior performance with fewer computational resources.

E. Ablation Study

In order to verify the effectiveness of the three components of ST2GCN, intra-subnetwork constraint and inter-subnetwork constraint in the proposed method, we conduct ablation experiments on epilepsy and ADNI datasets, and the experimental results are shown in Table IV and Table V. The baseline represents the classification of brain network using only MLP. We use $\checkmark$ to indicate that the component is used; Otherwise, it is not used. When comparing Scheme 1, Scheme 2, and Scheme 3 with the baselines, we observe that each component of our method, such as ST2GCN, intra-subnetwork constraints, and inter-subnetwork constraints, contributes to enhancing the performance of disease diagnosis. Comparing Scheme 4, Scheme 5 and Scheme 6 with Scheme 1, Scheme 2 and Scheme 3, we find that the performance of disease diagnosis improves when any two components are combined, than using only one component. We use the NC vs. FLE task as an example for quantitative analysis. Compared with baseline, adding ST2GCN, Intra-Subnetwork Constraint and Inter-Subnetwork Constraint modules can improve the performance by 6.6%, 4.0% and 2.5%, respectively. Furthermore, the combined use of ST2GCN and Intra-Subnetwork Constraint improves performance by at least 6.0% compared to using either module alone, the combined use of ST2GCN and Inter-Subnetwork Constraint improves performance by at least 7.0% compared to using either module alone, and the combined use of Intra-Subnetwork Constraint and Inter-Subnetwork Constraint improves performance by at least 8.9% compared to using either module alone. The experimental analysis shows that these modules are effective, and promote each other to improve the diagnostic performance. ST2GCN overcomes the shortcoming of traditional GCN that cannot deal with dynamic graphs, and can cooperatively extract the spatio-temporal features of DFBNs through intra-graph propagation and inter-graph propagation. The functional subnetwork constraint integrates functional and anatomical prior knowledge, so that the extracted spatio-temporal features can better capture the change in the topology of DFBN and are more discriminative.

V. DISCUSSION

A. Interpretability of Functional Subnetworks

Due to the incorporation of functional and structural priors of the brain, our proposed hierarchical DFBNs method inherently possesses interpretability. Through our proposed method, we elucidate the impact of different functional subnetworks on brain disease diagnosis from three respective aspects.

TABLE IV
THE CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON THE EPILEPSY DATASET (%)

Method	ST2GCN	Intra-Subnetwork Constraint	Inter-Subnetwork Constraint	ACC	F1	AUC	ACC	F1	AUC
				NC vs. TLE&FLE			NC vs. FLE		
baseline				69.3/05.0	73.8/07.0	63.4/08.8	69.0/08.0	63.9/09.4	59.9/10.9
Scheme 1	✓			77.4/06.4	82.6/05.3	72.9/12.1	75.6/05.9	64.3/15.4	74.8/09.7
Scheme 2		✓		74.2/05.1	80.0/05.0	71.3/11.2	73.0/09.3	61.2/19.8	71.7/11.7
Scheme 3			✓	73.2/04.9	79.5/04.8	64.5/12.8	71.5/07.4	61.3/16.8	67.5/13.5
Scheme 4	✓	✓		80.4/07.0	85.9/04.8	78.5/12.7	81.6/06.8	77.4/07.7	79.6/10.2
Scheme 5	✓		✓	78.8/07.3	84.0/07.1	72.0/14.5	82.6/06.1	76.7/07.9	84.9/04.4
Scheme 6		✓	✓	78.5/06.3	85.4/05.2	57.9/17.8	81.9/09.9	77.6/14.7	82.0/17.5
Ours	✓	✓	✓	84.0/05.0	88.1/04.4	82.0/06.7	83.5/05.5	80.2/05.5	87.1/06.0

Method	ST2GCN	Intra-Subnetwork Constraint	Inter-Subnetwork Constraint	ACC	F1	AUC	ACC	F1	AUC
				NC vs. TLE			TLE vs. FLE		
baseline				70.1/05.3	61.8/12.3	64.5/09.2	62.5/07.5	64.3/16.3	55.5/12.0
Scheme 1	✓			79.5/09.3	73.5/14.6	75.2/12.1	71.3/10.2	73.6/09.6	62.5/11.8
Scheme 2		✓		77.0/08.4	68.2/16.4	75.5/12.1	69.4/08.6	58.4/25.8	56.8/11.6
Scheme 3			✓	76.5/05.5	70.1/13.9	72.1/14.5	70.6/08.4	65.3/27.8	57.3/18.5
Scheme 4	✓	✓		79.9/08.2	75.1/13.4	80.0/17.4	71.3/08.5	68.9/24.0	68.3/10.5
Scheme 5	✓		✓	78.4/07.6	73.0/15.3	72.6/13.5	72.8/07.9	64.8/23.2	56.4/13.0
Scheme 6		✓	✓	78.0/08.1	73.0/11.9	74.0/10.3	73.1/10.1	67.8/25.1	61.4/20.0
Ours	✓	✓	✓	84.1/05.9	81.8/08.4	82.4/08.5	76.9/05.7	79.0/09.4	74.2/09.1

TABLE V
THE CLASSIFICATION RESULTS OF ABLATION EXPERIMENTS ON THE ADNI DATASET (%)

Method	ST2GCN	Intra-Subnetwork Constraint	Inter-Subnetwork Constraint	ACC	F1	AUC	ACC	F1	AUC
				NC vs. SMC&EMCI			NC vs. SMC		
baseline				77.9/07.8	81.7/08.1	76.3/10.5	83.0/07.8	77.7/11.1	84.5/12.0
Scheme 1	✓			85.0/04.9	87.7/04.6	83.4/12.2	86.0/05.6	82.8/09.1	88.4/06.6
Scheme 2		✓		83.2/04.9	85.5/08.0	78.9/08.8	83.9/10.6	78.0/17.8	83.4/10.8
Scheme 3			✓	80.1/04.4	84.2/04.7	79.3/08.4	84.8/05.8	80.3/09.8	82.4/09.5
Scheme 4	✓	✓		90.3/03.6	86.5/05.0	80.3/10.0	91.1/06.6	89.6/07.4	89.7/08.0
Scheme 5	✓		✓	86.3/08.6	87.4/10.0	88.4/06.3	89.5/06.5	87.7/06.7	89.9/07.4
Scheme 6		✓	✓	83.6/04.9	86.5/06.9	85.2/07.7	84.8/07.5	81.6/09.6	85.1/08.9
Ours	✓	✓	✓	93.4/02.2	95.1/02.3	92.5/06.5	93.7/05.0	93.0/05.1	95.3/04.8

Method	ST2GCN	Intra-Subnetwork Constraint	Inter-Subnetwork Constraint	ACC	F1	AUC	ACC	F1	AUC
				NC vs. EMCI			SMC vs. EMCI		
baseline				77.3/06.8	69.5/09.6	71.1/10.3	70.8/11.6	70.9/17.5	64.3/17.4
Scheme 1	✓			86.6/06.7	84.8/07.0	86.9/06.8	76.4/08.2	72.7/18.1	72.3/13.5
Scheme 2		✓		81.3/05.8	71.0/15.1	79.8/14.2	73.6/07.3	70.7/16.9	67.9/15.4
Scheme 3			✓	80.0/10.3	73.3/16.1	73.6/14.0	74.1/07.6	75.3/08.8	68.1/11.5
Scheme 4	✓	✓		90.7/06.8	87.3/11.0	90.6/10.5	88.9/09.4	89.7/08.9	89.9/08.3
Scheme 5	✓		✓	87.3/08.3	83.8/09.3	87.1/07.6	81.5/08.6	77.1/26.2	76.7/17.6
Scheme 6		✓	✓	87.3/07.5	84.7/08.6	86.0/10.1	75.8/07.4	74.6/10.0	67.5/12.8
Ours	✓	✓	✓	95.3/07.3	94.0/09.2	94.3/08.8	91.0/05.6	90.7/06.5	92.0/07.1

TABLE VI
CONTRIBUTION OF FUNCTIONAL SUBNETWORKS ON DIFFERENT
DIAGNOSTIC TASKS. EACH VALUE REPRESENTS THE IMPORTANCE OF
DIFFERENT SUBNETWORKS.[#]

Task	DMN	FPN	SAN	ATN	SMN	VN	AN	ON
NC vs. FLE	0.158	0.041	0.093	0.455	0.029	0.061	0.116	0.046
NC vs. TLE	0.335	0.060	0.046	0.277	0.035	0.170	0.027	0.049
NC vs. SMC	0.148	0.267	0.082	0.337	0.031	0.076	0.034	0.026
NC vs. EMCI	0.081	0.066	0.111	0.453	0.061	0.125	0.051	0.052

[#] We bold the two most important subnetworks for different tasks.

1) *The Significance of Each Functional Subnetwork*: We sum the attention weights obtained in Eq. (8) row-wise to obtain the significance of each subnetwork for diagnosis tasks. As can be seen from Table VI, the contributions of different functional

subnetworks to disease diagnosis are significantly different, which is caused by functional heterogeneity. Specifically, the two most important functional subnetworks in NC vs. FLE and NC vs. TLE tasks are DMN and ATN. This is because FLE stems from abnormal electrical activity in the frontal lobe [42], while TLE primarily arises from dysfunction in the temporal lobe [43], both of which are regions heavily represented in DMN and ATN. In the NC vs. SMC task, the two most important functional subnetworks are ATN and FPN. ATN and FPN are responsible for allocating attention and regulating cognitive resources to support complex cognitive tasks, respectively [44], [45]. Due to the occurrence of SMC, the brain need to allocate more cognitive resources to compensate for the defects in memory function. Additionally, memory problems can cause worry and anxiety in patients, resulting in difficulty

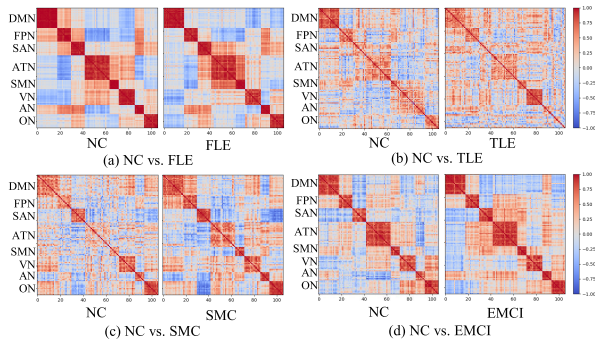


Fig. 5. Visualization of the extracted brain networks on different diagnosis tasks. Each pixel represents a brain connectivity value, with color intensity indicating connection strength. Warmer (redder) hues denote stronger connection weights.

for individuals to concentrate [46]. Therefore, SMC may partially arise from the abnormalities in ATN and FPN. In the NC vs. EMCI task, the two most important functional subnetworks are ATN and VN. EMCI can lead to cognitive decline, including memory ability, attention and visual perception, etc. ATN controls memory and attention, and VN determines an individual's visual and spatial perception ability. Therefore, our interpretable model can autonomously identify the significant subnetworks in different diagnosis tasks [45], [47].

2) Visualization of Brain Network: To visually compare the differences between brain networks of NC and patient groups, we visualized the average brain network of different groups in Figure 5. Specifically, we concatenate the spatio-temporal features of the eight functional subnetworks learned in the diagnosis task, and then use PC to calculate the brain network of NC and patient, respectively. As can be seen from Figure 5, the brain diseases lead to the occurrence of enhanced or weakened connectivity within functional subnetworks compared to the healthy group, which is caused by dysfunction and compensatory mechanisms [46], [47]. To further explore the alterations of intra-subnetwork connectivity caused by brain diseases, we calculated the average connectivity of the eight subnetworks. For NC vs. FLE, the average connectivity of all functional subnetworks is slightly higher in NC than that in the FLE group. For NC vs. TLE, we find that the connectivity of DMN, SAN, ATN, AN, and ON decreased in TLE patients compared to NC, but the connectivity of FPN, SMN, and VN increased. For NC vs. SMC and NC vs. EMCI, the connectivity of all subnetworks is significantly higher in SMC and EMCI patients than that in NC.

3) Inter-Subnetwork Connectivity: Studies have shown that the occurrence of brain diseases will not only cause abnormalities within functional subnetworks, but also change the information interaction between functional subnetworks [1]. Therefore, we calculate the inter-subnetwork connection strength of healthy control and patient groups on different classification tasks separately, and t-test is used to calculate the most significantly altered inter-subnetwork connectivities due to brain disorder. Figure 6 shows the two most significant inter-subnetwork connections on different classification tasks. For NC vs. FLE, we find that FPN-AN in FLE patients has higher connectivity strength than NC, while ATN-VN

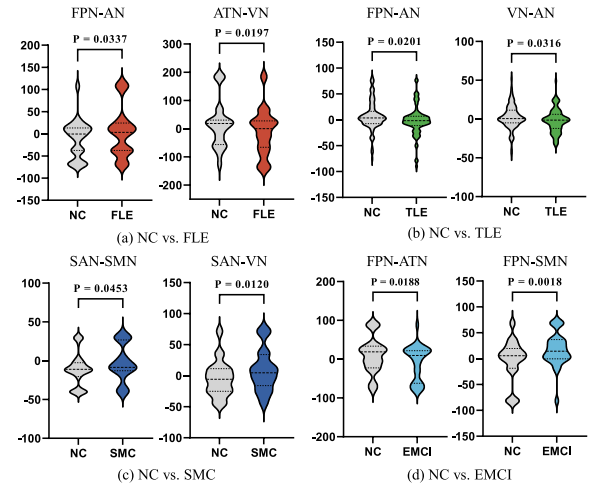


Fig. 6. Top two most significant inter-subnetwork connectivities on different diagnosis tasks. The vertical axis represents the connection strength between subnetworks.

has lower connectivity strength than NC. This is attributed to the cognitive deficits commonly observed in individuals with FLE, particularly affecting attention, spatial perception, and executive function [48]. The FPN, AN, ATN, and VN networks play pivotal roles in cognitive control, attention, and visual processing within the brain [1]. Consequently, FLE can disrupt the normal transmission of information among these subnetworks, potentially resulting in aberrant cognitive function. For NC vs. TLE, we find that the connectivity strength of FPN-AN and VN-AN is significantly lower in TLE patients than that in NC, due to the impairment of episodic memory, visual and auditory functions in TLE patients, which inhibits the interaction between the corresponding functional subnetworks [43], [49]. For NC vs. SMC, we find that the connectivity strength between SAN-SMN and SAN-VN is significantly higher in SMC patients than that in NC. SAN, SMN, and VN are responsible for adaptive cognitive response, motor, and visual functions, respectively [1]. The occurrence of SMC will cause the decline of cognitive ability and physical control ability, so the brain may enhance the connectivity between SAN-SMN and SAN-VN to compensate for these functional defects [50]. For NC vs. EMCI, we find that the FPN-ATN connectivity strength is lower in EMCI patients than that in NC, while the FPN-SMN connectivity strength is higher in NC. This is because the cognitive ability of EMCI patients is further decreased compared with that of SMC patients, which leads to the weakened connectivity of phase FPN-ATN. In addition, in order to ensure the normal somatic control function of EMCI patients, the connectivities of related functional subnetworks such as FPN-SMN are enhanced [51].

B. Spatio-Temporal Variability of DFBNs

In this subsection, we calculate the variance of the total connectivity strength of the extracted brain network in all sliding windows. We consider variance as the spatio-temporal variation rate and explore the influence of brain diseases on the spatio-temporal characteristic of DFBNs. The experimental results are shown in Figure 7. From Figure 7, we can find that

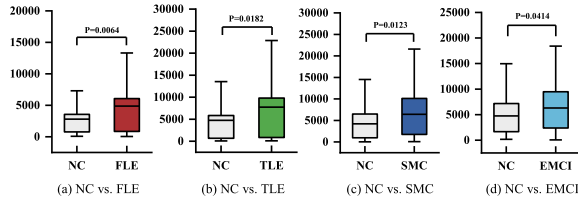


Fig. 7. Comparison of spatio-temporal variability of DFBNs on different classification tasks. The vertical axis represents the spatio-temporal variability of DFBNs.

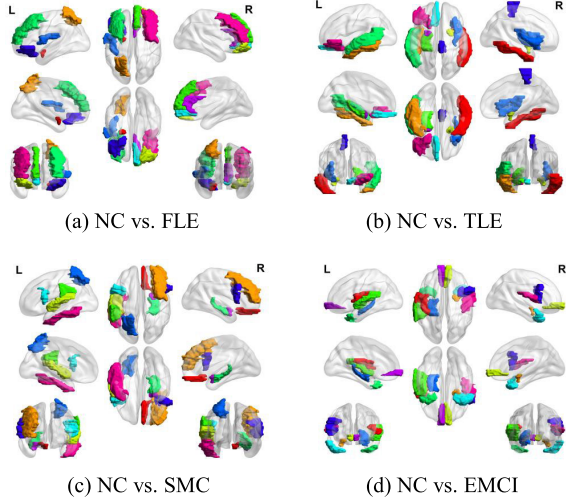


Fig. 8. Distribution locations of the 10 most discriminative brain regions on different diagnosis tasks.

the spatio-temporal variation rate of patients is significantly higher than NC, which indicates that brain diseases can cause the brain connectivity state to become unstable. Epilepsy is caused by the excessive electrical activity of neurons caused by the disorder of the inhibition mechanism of neurons [42]. This abnormality will spread to other areas of the brain network, resulting in the interruption or enhancement of connectivities, making the changes in brain connections more drastic. SMC and EMCI are cognitive disorders, in which patients' cognitive abilities such as memory ability and attention will gradually deteriorate, and neurodegenerative changes will occur in the brain. In addition, the brain undergoes network remodeling to compensate for these dysfunctions, so brain networks become more unstable. The similar conclusions are also mentioned in literatures [50], [52] and [53].

C. Discriminative Brain Regions

We use t -test to find the most important (P -value < 0.1) brain regions identified by the proposed method, and the experimental results are shown in Figure 8. For NC vs. FLE, the discriminative regions include Amygdala.L, Frontal.mid.orb.R, Frontal.sup.medial.R, and Middle frontal gyrus, which are responsible for regulating cognitive and behavioral functions. They play an important role in the diagnosis of FLE [48]. For NC vs. TLE, the discriminative regions included Temporal.mid.L, Amygdala.R, Hippocampus.L, and Inferior temporal gyrus. We can find that the sought biomarkers correspond to regions associated with functional deficits such as memory loss, language or visual impairment in TLE patients [43],

[49]. For NC vs. SMC, discriminative regions include Frontal.mid.R, Temporal.sup.L, Hippocampus.R and Inferior frontal gyrus, etc., which can effectively distinguish between the NC and SMC patients [46], [54]. For NC vs. EMCI, discriminative regions are concentrated in Rolandic.Oper.L, Amygdala.R, Hippocampus.L, and Temporal.Pole.Mid, which have been shown to be useful in discriminating EMCI [46]. Therefore, our method can provide effective biomarkers for epilepsy and cognitive impairment diagnosis.

VI. CONCLUSION

In this paper, we propose a hierarchical representation model for DFBN analysis integrating functional and anatomical priors in the brain, and develop the ST2GCN to collaboratively extract dynamic topological features from hierarchical DFBNs. Specifically, we first construct hierarchical DFBNs with functional and structural prior. Secondly, we develop a graph convolution network with intra-graph and inter-graph propagation to collaboratively extract the dynamic topological features within and among subnetworks. Then, considering the brain functional and structural clustering prior, we introduce a functional subnetwork constraint in the proposed ST2GCN, which is devised to amplify the uniformity of expression within each subnetwork while accentuating expression disparities between subnetworks. Finally, we employ the self-attention mechanism to adaptively fuse the spatio-temporal features of different subnetworks to obtain the global representation for classification. Experimental results on epilepsy, ADNI and ABIDE datasets show that our method outperforms state-of-the-art brain network analysis methods. Moreover, it inherently offers interpretability by leveraging embedded prior knowledge of brain structure and function, thus providing hierarchical biomarkers for brain disorders.

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